KINEMATIC AND KINETIC PARAMETERS OF THE SPRINT START AND START ACCELERATION MODEL OF TOP SPRINTERS

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The purpose of the research was to find the most important kinematic and kinetic parameters of the set position and starting action and their correlation with the start acceleration. The subject sample comprised of thirteen male sprinters and eleven female sprinters. A 2D kinematic analysis video system (APAS) was used to register the kinematic start and start acceleration parameters. The kinetic parameters of the starting action were measured with the help of modified starting blocks (MMIP) with in-built measurement sensors. The time parameters of the start acceleration were measured by four pairs of photo cells (AMES) placed at 5–10–20–30 m from the starting line. The statistical analysis was performed with the statistical package SPSS. The efficiency of the sprint start for both sexes is generated by: horizontal start velocity of the C. G., starting reaction time, force impulse and the maximal force gradient on the front starting block. Significant differences ($p < 0.05$) between the genders are predominantly in the area of kinetic parameters of the starting action. The highest correlations with the start acceleration for male sprinters has the kinetic parameters block of the starting action – maximal and relative force of pressure, maximal force gradient, force impulse and time to maximal force; among the kinematic parameters horizontal start velocity of the C. G. and the ankle angle in the front starting block. Female sprinters have much lower correlations, only two significant coefficients were obtained – time to maximal force on the front and rear starting block. The generally low correlations between the start and the start acceleration, especially in the first five metres from the start are the consequence of the differences in the biomechanical motor structure.

Keywords: sprint start, start acceleration, kinematics, kinetics, top sprinters, males, females.

INTRODUCTION

The start and the start acceleration are two most important phases, directly influencing the final result in sprint. It is therefore not surprising that numerous biomechanical studies have been carried out in the last few years and some very important information obtained, both for theoretical and practical use in the training process of sprinters (Hoster, 1981; Mero, Luhtanen & Komi, 1983; Moravec, Ruzicka, Susanka, Dostal & Nosek, 1988; Mero, 1988; Coppenolle, Delecluse, Goris, Bohets & Eynde, 1989; Inglis, 1989; Coppenolle, Delecluse, Goris, Seagrave & Kraayenhof, 1990; Guissard, Duchatenau & Hainaut, 1992; Delecluse, Coppenolle, Diels & Goris, 1992; Korchemny, 1992; Schot & Knutzen, 1992; Taylor, 1994; McClements, Sanders & Gander, 1996). With the advancement of modern measurement technology new possibilities appear of finding relevant and more objective information on the execution of the sprint start and start acceleration in actual training or competitive conditions.

The aim of our research was to determine the most important kinematic and kinetic parameters of the "set" position and starting action, as well as their correlation with the start acceleration, defined by time points at 5–10–20 and 30 m.

According to the aim of the research the following two hypothesis can be formulated:
H1: Male and female sprinters differ both in the kinematic as well as the kinetic parameters of the set position and starting action.
H2: Kinematic and kinetic parameters of the start position and start action are most highly correlated with the initial phase of the start acceleration (0–5 m), both for male and female sprinters.

METHOD

Subjects

The subject sample consisted of thirteen male sprinters and eleven female sprinters of the Slovene national selection. The average age of the male sprinters was 24.4 ± 3.1 years, average height 176.7 ± 5.1 cm and average weight 76.7 ± 6.5 kg. The average 100 m result was 10.73 ± 0.2 s, the best 10.49 s. For the female sprinters the averages were: age 23.6 ± 2.9 years, height 165.1 ± 2.7 cm and weight 57.0 ± 2.5 kg. The best 100 m result was 11.55 s and the group average 11.97 ± 2.6 s.

Measurement protocol and technology

The measurements were executed before the 1996 summer competition season. The measurment was performed indoor in order to ensure maximally constant conditions. Each subject performed three sprints from the starting blocks on a distance of 30 m. A ten minute break was prescribed between the individual attempts. The placement of the starting blocks was individually set according to the wishes of the athletes. Standard starting commands were used. The best achieved result in the 30 m sprint of each individual was used in the study.

The measurement system consisted of three units (Fig. 1): starting blocks (MMIP) with computer support, the kinematic system APAS (Ariel Performance Analysis System) and an electronic measurement system with four pairs of photo cells (AMES) for measuring the time parameters of the start acceleration.

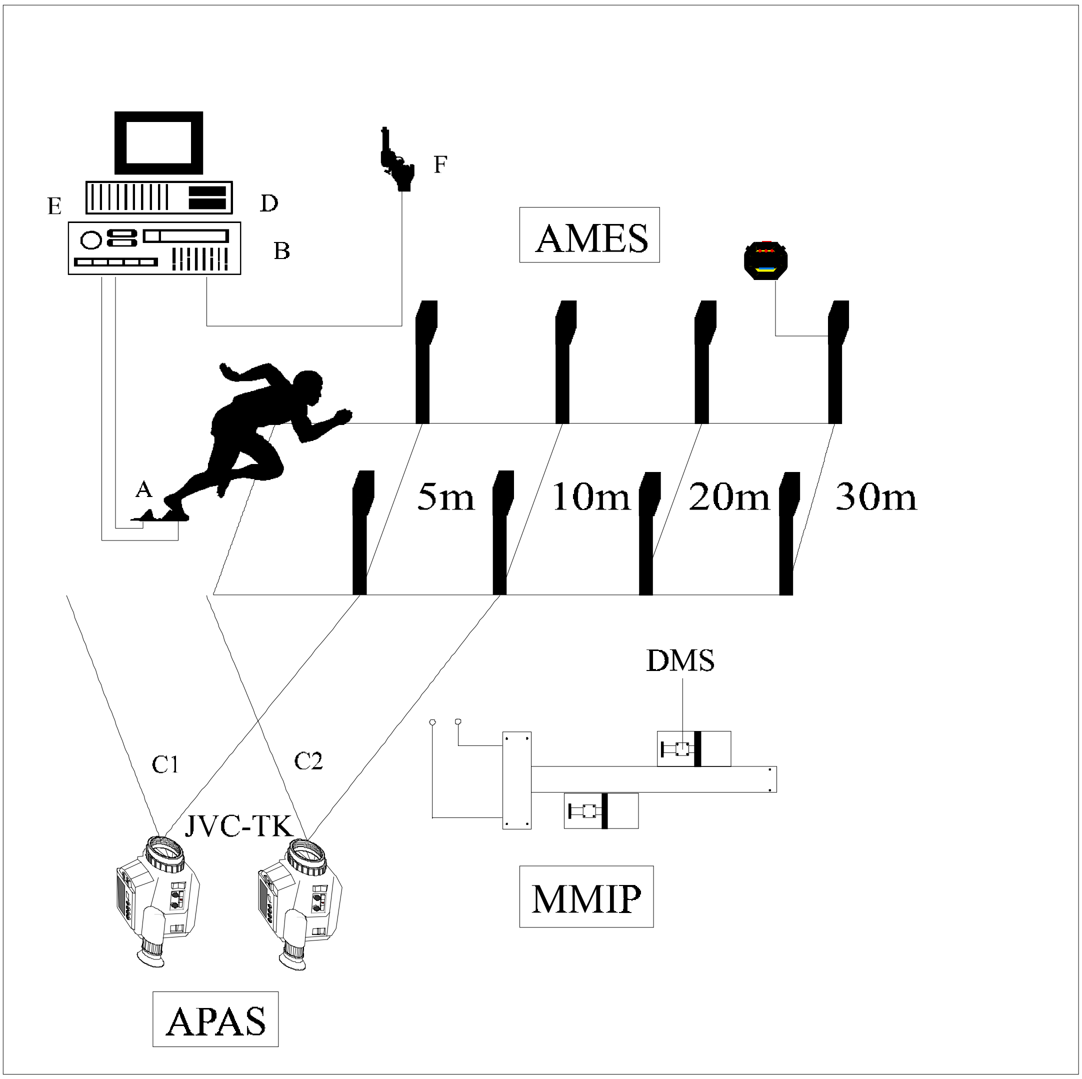

Fig. 1
Measuring system for measuring kinematic and kinetic variables of the sprint start and starting acceleration. (A) modified starting block – MMIP; (B) amplifier MES HPSC 3102; (D) computer PCX 486; (E) AD converter; (F) starting gun; (APAS) kinematic system; (AMES) timing system

A modified starting block MMIP, developed in the Laboratory of Biomechanics of the Faculty of Sport in Ljubljana together with the Faculty of Natural Sciences and Technology in Ljubljana, was used to assess the kinetic parameters of the start. Measurement sensors of aluminium alloy with glued on strain-gauges (DMS) were built into the front and rear starting block. The strain-gauges are interconnected in the form of a Wheatstone bridge and work on the principle of change of resistance resulting from the external strain forces on the measurement sensors. The strain force can have the range from 50 N to 2500 N. The measurement sensors register the horizontal component of the force of pressure on the blocks, with the signal going to the amplifier and then to the converter. The computer unit reads the data with a frequency of 1000 Hz. The analysis of the data and the force diagrams on the front and rear block (Fig. 2) was performed with the computer programme package BITC-11 (Bizjak, Tercelj & Fajfar, 1993).

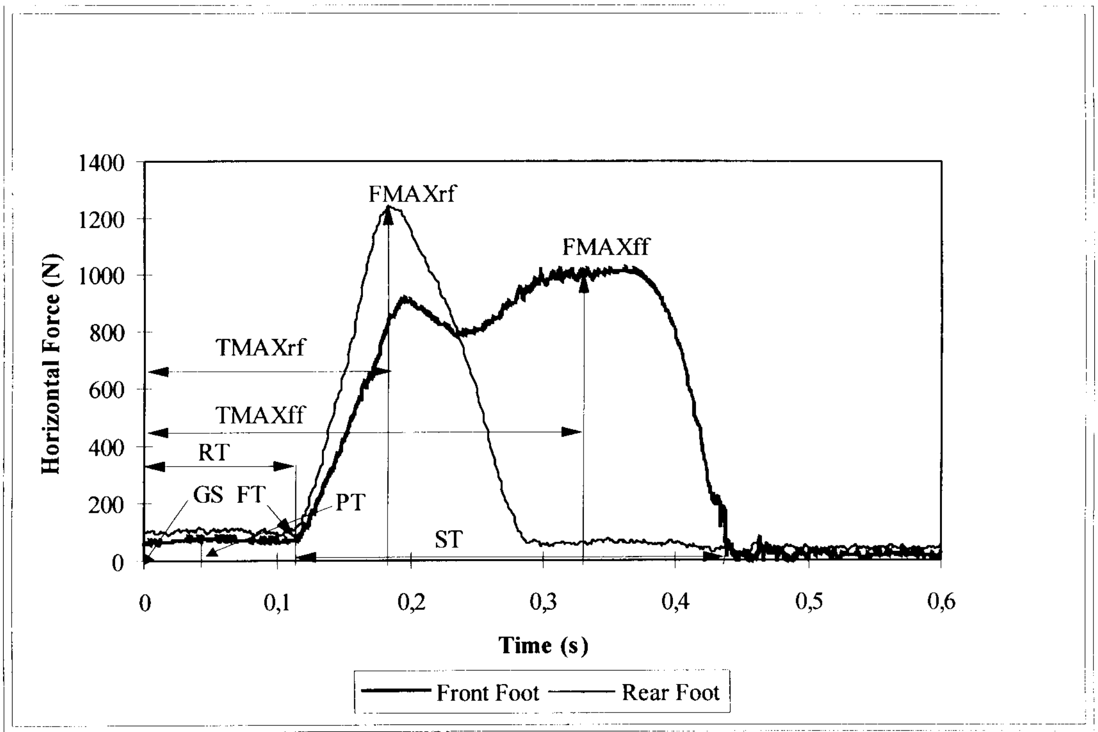

Fig. 2
Diagram of force development in the starting action. (GS) gun start; (PT) pretension; (RT) latent reaction time; (FT) force treshold; (FMAXrf) maximal force of rear foot; (FMAXff) maximal force of front foot; (ST) starting time; (TMAXrf) time till maximal force of rear foot; (TMAXff) time till maximal force of front foot; (ZISO) area under curve of rear foot; (SISO) area under curve of front foot

The origin of the co-ordinate system (xy = 0) is defined by the starting pistol signal. The pretension (PT) with which the athlete presses with the foot on the block in the horizontal plane lasts till point FT. This is the starting point, when the force of pressure surpasses the 10 % of maximal force treshold and signifies the beginning of the starting time phase. The time interval from the starting pistol signal to point FT is the reaction time RT.

The kinematic parameters were assessed by a two-dimensional video system for kinematic analysis APAS (Ariel Performance Analysis System). The set position, starting action and start acceleration (Fig. 1) were filmed with two synchronised SVHS cameras JVC TK-1281 EG with a frequency of fifty frames per second – placed perpendicular to the running direction. The first camera (C1) registered the movement of the starting action, while the second camera (C2) encompassed the movement of the sprinter during the start acceleration up to the distance of 10 m. The room was spanned with a reference cube (100 cm x 100 cm x 100 cm) and defined by X – the horizontal axis and Y – the vertical axis.

In further procedure a digitalisation was performed of a 16-segment model of the athlete's body (head, shoulders, upper-arm, lower-arm, trunk, thigh, shank, foot – the "Susanka" model) defined by the co-ordinates of eighteen reference points. Co-ordinate filtering was performed with the help of a Cubic-Spline filter. The data that issued from these procedures was used to compute the relevant kinematic parameters of the start and first two strides of the start acceleration. The time parameters of the start acceleration were measured with a measurement system AMES, using four pairs of photo cells placed at 5–10–20–30 m.

The SPSS statistical package was used for statistical data analysis. Basic statistics of the variables was computed, T-test was used to assess the significance (on the 5 % alpha error level) of the differences between the male and female sprinters in the parameters of the start and start acceleration. The correlation between the parameters was assessed with Pearson correlation coefficients.

RESULTS

The statistics of the kinematic parameters (TABLE 1, Fig. 3) shows important information on the position of the individual segments of the body, in light of the placement of the starting blocks in the set position. For male sprinters the front block was placed 55.1 ± 6.2 cm (females 45.5 ± 5.4 cm) from the starting line, the rear block 81.9 ± 74 cm (females 68.9 ± 4.6 cm). According to a study (Schot & Knutsen, 1992) this is a „medium anteroposterior" position of the blocks. The height of the centre of gravity (C.G.) is 54.4 ± 4.8 cm for male sprinters and 53.2 ± 2.0 cm for the female sprinters, the horizontal distance of the projection of C.G. from the starting line is 18.8 ± 5.1 cm (males) and 15.7 ± 3.0 cm (females) respectively.

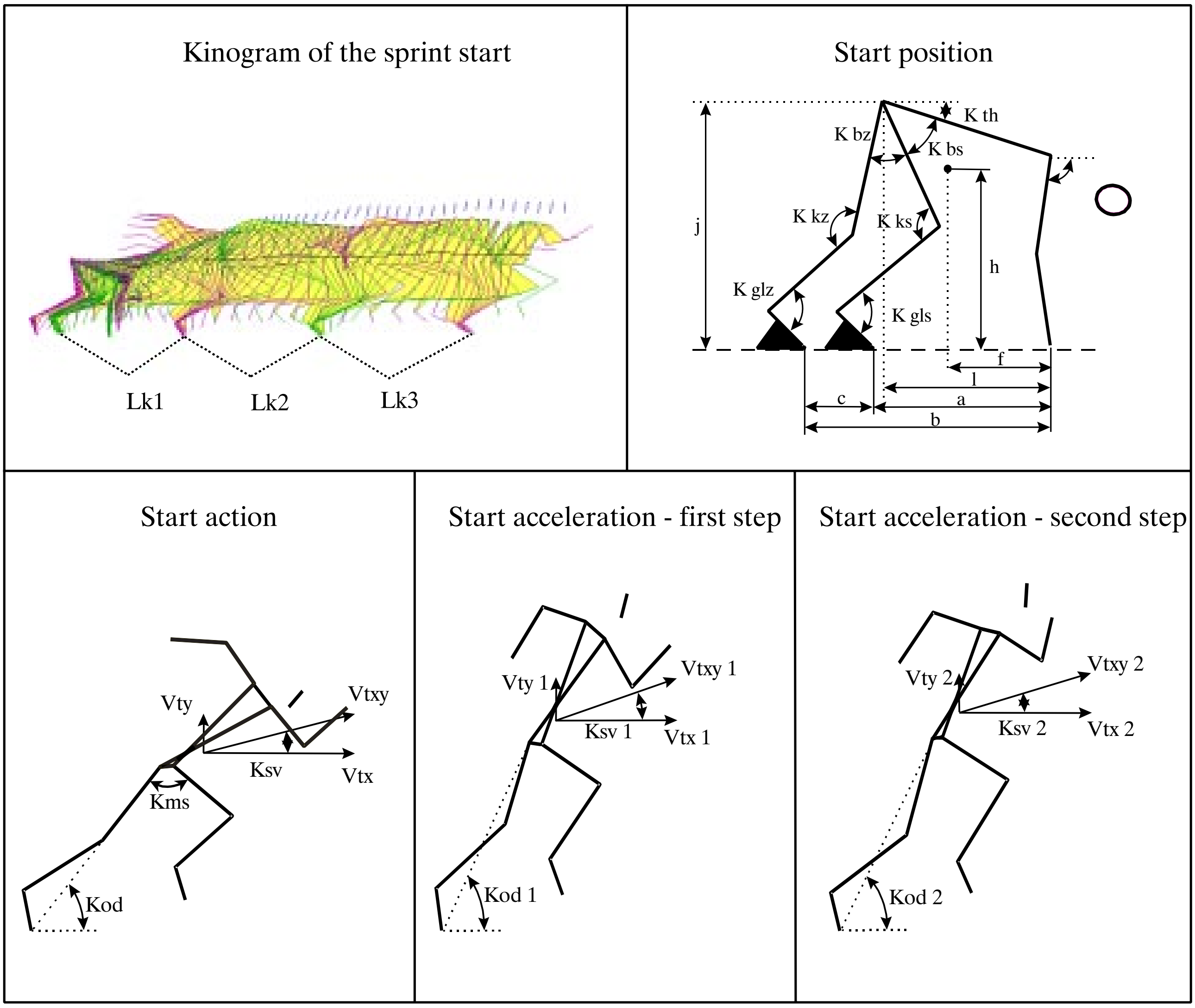

Fig. 3
Kinematic variables of the start and start acceleration

An important parameter of the set position is the angle in the knee of the rear leg, here it was 112 ± 13.3° for male sprinters and 115 ± 13.8° for female sprinters. A greater value of the knee angle of the front and rear leg of female sprinters results in the absolute height of the hips – the male and female athletes were very similar in this regard, the difference was namely only 1.22 cm.

Table 1
Basic statistics and T-test of kinematic variables of the set position, starting action and acceleration

VARIABLE		MALES		FEMALES		T-TEST	
START POSITION	**U**	**M**	**SD**	**M**	**SD**	**T**	**SIG**
Distance front block-starting line (a)	**cm**	55,15	6,22	45,49	5,37	4,03	0,00**
Distance rear block-starting line (b)	**cm**	81,88	7,44	68,96	5,91	4,65	0,00**
Distance between the blocks (c)	**cm**	26,72	2,33	23,47	1,88	3,71	0,00**
Absolute height of C.G. (h)	**cm**	54,38	4,81	53,18	2,04	0,08	0,42
Relat. height of C. G.-height of C. G.: body height (g)		0,30	0,02	0,32	0,02	–2,22	0,04*
Projection of C. G. from starting line (f1)	**cm**	18,77	5,07	15,73	3,00	1,74	0,95
Absolute height of hips (j)	**cm**	72,31	7,71	71,09	3,33	0,52	0,61
Relat. height of hips-height of hips: body height (k)		0,41	0,04	0,43	0,02	–1,93	0,07
Projection of hips position from starting line(l1)	**cm**	37,31	7,30	33,45	4,06	1,63	0,12
Ankle angle of front leg (Gls)	**°**	97,55	10,55	102,65	6,58	–1,39	0,18
Ankle angle of rear leg (Glz)	**°**	97,45	10,28	99,80	6,44	–0,65	0,52
Knee angle of front leg (Kks)	**°**	93,75	8,26	103,38	6,97	–3,05	0,01*
Knee angle of rear leg (Kkz)	**°**	112,72	13,31	115,59	13,86	–0,52	0,61
Angle in hips: trunk-front leg (Kbs)	**°**	44,78	6,15	42,36	9,43	0,75	0,46
Angle in hips: trunk-rear leg (Kbz)	**°**	24,91	4,27	19,25	9,30	1,86	0,09
Position of trunk: trunk-horizontal (Kth)	**°**	20,05	8,60	24,20	5,12	–1,46	0,16
Position of arms: upper arm-horizontal (Knh)	**°**	73,26	6,48	74,07	7,71	–0,28	0,78

STARTING ACTION							
Start push-off angle (Kod)	**°**	49,54	2,91	53,2	3,2	–2,93	0,01*
Vertical start velocity (Vty)	**ms^{-1}**	0,69	0,21	0,76	0,19	–0,73	0,48
Horizontal start velocity (Vtx)	**ms^{-1}**	3,2	0,19	2,99	0,23	2,48	0,02*
Start velocity resultant (Vtxy)	**ms^{-1}**	3,37	0,35	3,09	0,21	2,34	0,03*
Angle of start velocity resultant (Ksv)	**°**	12,12	4,21	14,34	4,06	–1,31	0,21
Angle between the thighs (Kms)	**°**	57,02	12,54	52,54	19,44	0,68	0,50
START ACCELERATION – FIRST STRIDE							
Lenght of first stride (Lk1)	**cm**	100,85	9,79	98,64	6,74	0,63	0,53
Push-off angle (Kod 1)	**°**	56,65	3,50	58,67	3,11	–1,49	0,15
Vertical velocity (Vty 1)	**ms^{-1}**	0,37	0,19	0,52	0,10	–2,18	0,04*
Horizontal velocity (Vtx 1)	**ms^{-1}**	4,47	0,29	4,25	0,18	2,20	0,04*
Velocity resultant (Vtxy1)	**ms^{-1}**	4,49	0,29	4,29	0,18	2,00	0,06
Angle of velocity resultant (Ksv 1)	**°**	4,78	2,16	6,92	1,44	–2,80	0,01*
START ACCELERATION – SECOND STRIDE							
Lenght of second stride (Lk2)	**cm**	103,77	8,96	107,18	6,23	–1,06	0,30
Push-off angle (Kod 2)	**°**	59,93	2,79	64,13	4,44	–2,82	0,01
Vertical velocity (Vty 2)	**ms^{-1}**	0,45	0,18	0,5	0,10	–0,83	0,42
Horizontal velocity (Vtx 2)	**ms^{-1}**	5,38	0,24	4,99	0,29	3,63	0,00**
Velocity resultant (Vtxy 2)	**ms^{-1}**	4,40	0,24	5,01	0,29	3,6	0,00**
Angle of velocity resultant (Ksv 2)	**°**	4,89	1,73	5,79	1,23	–1,44	0,16
START ACCELERATION							
Time to 5 m (c5)	**s**	1,29	0,04	1,39	0,03	–8,14	0,00**
Time to 10 m (c10)	**s**	1,98	0,04	2,13	0,02	–12,76	0,00**
Time to 20 m (c20)	**s**	3,13	0,06	3,38	0,04	–14,42	0,00**
Time to 30 m (c30)	**s**	4,19	0,07	4,53	0,07	–13,40	0,00**

Legend:
U – measurement unit
M – arithmetic mean
SD – standard deviation
T – test
SIG – significant off t-test
*** p < 0,05**
**** p < 0,01**

The male and female sprinters differ significantly only in five of the parameters of the set position according to the T-test statistics, that is in the placement of the starting blocks (a, b, c), relative height of the hips (g) and the knee angle of the front leg (Kks).

Table 2
Basic statistics and T-test of kinetic variables of the start action

		FRONT FOOT						**REAR FOOT**					
		MALES		**FEMALES**		**T-TEST**		**MALES**		**FEMALES**		**T-TEST**	
VARIABLE	**U**	**M**	**SD**	**M**	**SD**	**T**	**SIG**	**M**	**SD**	**M**	**SD**	**T**	**SIG**
Relative push-off force impulse (IRSO)	**Ns/kg**	2,6	0,51	2,45	0,56	0,67	0,51	1,3	0,38	1,33	0,42	–0,52	0,61
Push-off force impulse (ISO)	**Ns**	201,4	49,01	140,41	31,85	4,00	0,00**	97,6	32,19	76,1	23,62	2,04	0,05*
Maximal force gradient (MGS)	**N/s**	16484,2	4143,86	14410,38	6186,96	1,07	0,30	26896,1	8648,02	20346,79	6281,95	2,32	0,03*
Starting reaction time (ST)	**s**	0,3	0,03	0,34	0,02	–1,95	0,06	0,2	0,02	0,18	0,03	–0,99	0,33
Max. relative force of pressure (MRSP)	**N/kg**	12,1	2,31	12,04	3,21	0,01	0,99	12,9	3,45	12,54	2,39	0,29	0,77
Maximal force of pressure(MSP)	**N**	936,2	205,74	688,6	175,19	3,48	0,00**	1002,6	287,43	719,76	137,65	3,41	0,00**
Average force gradient (PGS)	**N/s**	3246,0	1009,18	2132,22	625,46	3,60	0,00**	12734,0	4245,1	7319,45	1942,24	4,46	0,00**
Average relative force gradient (PRGS)	**N/kg/s**	41,7	11,29	37,23	11,10	1,08	0,29	163,4	50,36	127,06	32,75	2,32	0,03*
Average relative force of pressure (PRSP)	**N/kg**	7,9	1,26	7,22	1,45	1,44	0,16	7,2	1,82	7,28	1,52	–0,08	0,94
Average force of pressure (PSP)	**N**	619,2	130,24	414,44	85,02	5,04	0,00**	563,2	152,03	417,62	86,89	3,19	0,00**
Latent reaction time (RT)	**s**	0,1	0,01	0,14	0,03	–0,17	0,11	0,1	0,01	0,13	0,03	–1,38	0,19
Time to maximal force (TMAX)	**s**	0,4	0,03	0,42	0,04	–3,97	0,00**	0,2	0,02	0,22	0,04	–2,66	0,02*

Legend:
U – measurement unit
M – arithmetic mean
SD – standard deviation
T – test
SIG – significance of T-test
*** p < 0,05**
****p < 0,01**

Both kinematic (TABLE 1) as well as kinetic parameters (TABLE 2) exert influence on the execution of the starting action. The horizontal start velocity which defines the efficiency of the first phase of the start acceleration was 3.20 ± .19 ms^{-1} for male sprinters and 2.99 ± .23 ms^{-1} for female sprinters. The starting time tells us how much time the athlete needs for the push-off phase from the starting blocks. This was 0.30 ± .03 s for males and 0.34 ± .02 s for females. However, the efficiency of the start execution is not dependent just on the starting time but also on the push-off force from the blocks. A criterion of this action is the impulse, measuring 201 ± 49 Ns on the front block and 97 ± 32 Ns on the rear block for males. Male sprinters manage to achieve a maximal force of 936 ± 205 N on the front block (female sprinters 688 ± 175 N) and 1002 ± 287 N (female sprinters 719 ± 137 N) on the rear block.

In comparison with the kinematic variables, we see a greater number of variables statistically significantly differentiating the two subsamples in the kinetics of the starting action. The differences are therefore to a greater extent the consequence of the different genetic potential of power than of the technical execution of the start.

Theefficiency of the transition from the start to the start acceleration is shown mostly by the parameters of the velocity of C.G. in the first two strides (TABLE 1, Fig. 3). The horizontal velocity of C.G. at the end of the first stride (rear support phase) is 4.47 ± .29 ms^{-1} and 5.38 ± .24 ms^{-1} at the end of the second stride. The horizontal velocity of male sprinters has therefore increased by 2.18 ms^{-1} from the moment of leaving the front block to the rear support phase of the second stride. This figure is 2.00 ms^{-1} for female sprinters.

The criterion of start acceleration is the velocity developed by the athletes in the start to 30 m segment. Male sprinters achieved a velocity of 3.87 m/s in the first 5 m, 5.05 m/s at 10 m, 6.38 m/s at 20 m and 7.16 m/s at 30 m. The female sprinters' velocity development was as follows: 3.59 m/s at 5 m, 4.69 m/s at 10 m, 5.91 m/s at 20 m and 6.62 m/s at 30 m (Fig. 4).

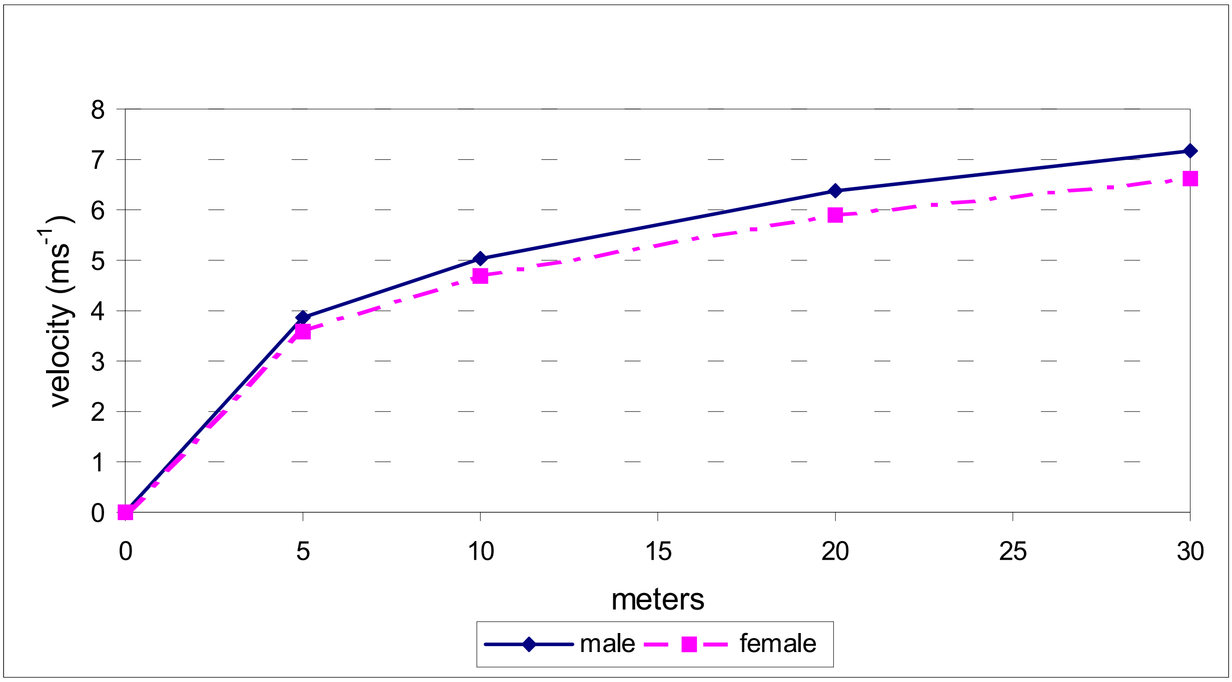

Fig. 4
Start acceleration of male and female sprinters

DISCUSSION

The set position in the blocks is very individual and depends mainly on the athlete's morphologic characteristics and motor abilities. Specially important are the co-ordinates of C.G. in the X and Y axes in regard to the starting line. The height (Y) of C.G. is 54.3 cm for male sprinters, being 31 % of the standing height – identical to the result found by A. Mero (1988). The horizontal distance of the C.G. from the starting line (X) is 18.8 cm, which is 11 % of the standing height of the sprinter. For female sprinters the height of the C.G. in the set position is 32 % and the horizontal distance of the C.G. from the starting line 9 % of the standing height. This leads us to the conclusion that female sprinters have their C.G. closer to the starting line, manifesting itself in the greater angle in the shoulders in regard to the horizontal (Knh) and a significantly greater knee angle of the front leg (Kks). A research (Schot & Knutzen, 1992) showed that blocks placed further from the starting line enable a greater horizontal starting velocity, but at the same time the authors did not obtain a positive effect of the set position, where the C.G. is very close to the starting line, on the efficiency of the start acceleration.

The latent reaction times (RT) are similar to those found by other authors (Mero, 1988; Moravec et al., 1988; Coppenolle, 1989). No significant differences exist between male and female sprinters in this ability. Male sprinters' reaction of the front foot was 0.03 s faster than that of the rear foot, for females the difference was 0.01 s. This parameter is limited with the value 0.10 s, a lower value is according to the international track and field regulations a false start.

A significantly more important role in the execution of the start goes to the start reaction time (ST) where the athlete has to build up as high a force on the starting blocks in as short a time as possible. According to Coppenolle (1992) this parameter is, besides horizontal start velocity and horizontal start acceleration, the most important factor of start efficiency. The differences between the male and female sprinters in the start reaction time of the front foot are on the verge of statistical significance.

In TABLE 2 are shown some of the most important kinetic variables of the starting action. The average value of the force impulse (IF = force x time) in the horizontal direction on the front block is 201 Ns for our male sprinters, which is less than found by Coppenolle (1989) for top sprinters. For female sprinters however, we found greater values of the force impulse on the first block (140 Ns) than the above-mentioned author. The relative force impulse (impulse : body weight = 2.45 ± 0.56 Ns/kg) is also greater than the one given by Coppenolle in his study (1989) for some top female sprinters (S. Echols, M. Ottey, N Cooman).

Among the kinetic variables significant differences exist between the male and female sprinters in maximal force on the blocks. This difference amounts to 248 N on the front block and 283 N on the rear block. This finding is not surprising since the amount of muscular mass is also one of the generators of maximal force. The gender differences become non-significant when we standardise force with body mass.

The two groups differ significantly also in the force gradient on the front and the rear block. The force gradient (change of force in a unit of time) is a criterion of the level of explosive power of the sprinter (Bührle, 1983).

The level of correlation between the kinetic and kinematic variables of the set position and starting action can be assessed through their inter-correlation coefficients. The criterion of the efficient start is the horizontal start velocity (Coppenolle, 1989; Delecluse, 1996). This parameter has significant correlation coefficients with the force impulse from the front block ($r = .54$), reaction time of front foot ($r = -.57$) and the ankle angle of the front ($r = -.62$) and rear leg ($r = -.53$). The significant negative correlation coefficients of the ankle angle of the front and rear leg show that efficiency of the horizontal start velocity is connected with dorsal flexion (a sharp angle between the foot and the shank) of the leg in the starting blocks. The average value of the ankle angle of the front leg is 97.5° and 97.4° of the rear leg. The effect of a greater obliquity of the front block on start velocity was studied by Guissard (1992) through EMG activity of the gastrochnemius, soleus and vastus medialis. She found that greater obliquity of the front block (30°–50°) statistically significantly effects greater start velocity, without prolonging the push-off phase from the block. The increase in start velocity is the consequence of aizometric-concentric muscular contraction of the back shank muscles and the use of the elastic energy in the concentric phase. Besides the muscles themselves, the muscle spindles also extend in the excentric phase, causing a reflex activity that raises EMG in the muscle. The obliquity of the block causes both neural as well as mechanical changes in the functioning of the ankle extensors. In the position of dorsal flexion the Achilles tendon also stretches to some extent, storing some elastic energy in the tendon (Bosco, Komi & Ito, 1981; Bosco, 1985) which is freed in flexion. The greater the extension the greater the work the tendon will perform. The efficiency of the push-off from the blocks is therefore the consequence of the action of two energy systems – chemical energy of the extensors of the ankle and the elastic energy of the connective and elastic elements.

With the female sprinters the horizontal start velocity is statistically significantly correlated with the parameter of the set position – knee angle of the rear leg ($r = -.62$) and parameters of the starting action – start push-off angle ($r = -.76$) and the maximal angle between the thighs ($r = .79$). The push-off from the blocks under a oblique angle ($M = 53.2 \pm 3.2°$) ensures a lesser vertical and a greater horizontal component of velocity of the C.G. The efficiency of the starting action depends mostly, according to some studies (Mero, 1988; Coppenolle, 1989; Sanderson, 1991; Schot, 1992), on the horizontal start velocity and starting time – resulting in the horizontal start acceleration.

The highest correlations with the horizontal start velocity have: the maximal force gradient on the front block ($r = .85$), maximal force of pressure of the foot on the front block ($r = .74$), average relative force gradient on the rear block ($r = .65$), time to maximal force on the front block ($r = .89$) and the rear block ($r = .78$). In light of the significance of the coefficients we can state that both kinetic and kinematic parameters define the quality of the start for both groups. However, with male sprinters the key ones are the kinetic parameters of the front block, while for female sprinters it is the kinetic parameters of the front and rear block, as was also found by other authors (Delecluse, 1992; McClements, 1996).

The correlation between the kinematic and kinetic parameters of the start with the start acceleration, defined with the times to 5–10–20–30 m, can be seen from the correlation coefficients in TABLE 3. The most marked correlation for male sprinters is between the group of kinetic parameters of the start with the results of the start acceleration at 20 and 30 m. The start acceleration at 30 m is statistically significantly correlated with all the kinematic start variables, without exception. For female sprinters the situation is quite different, as only two significant coefficients were obtained – the time to maximal force on the front (r = .61) and the rear block (r = .69).

Table 3
Correlation analysis of the chosen kinematic and kinetic variables of the start with start acceleration

		MALES				FEMALES			
		CORRELATION				CORRELATION			
VARIABLE	U	C5	C10	C20	C30	C5	C10	C20	C30
SET POSITION									
Ankle angle of front leg (Gls)	°	0,53	0,53	0,63*	0,40	0,30	0,33	0,09	0,03
Ankle angle of rear leg (Glz)	°	0,50	0,43	0,53	0,36	0,21	–0,0007	–0,11	–0,37
Knee angle of front leg (Kks)	°	0,30	0,29	0,50	0,35	–0,28	0,44	0,54	0,69*
Knee angle of rear leg (Kkz)	°	0,00	–0,01	0,23	0,11	0,13	0,54	0,57	0,30
Angle in hips: trunk-front leg (Kth)	°	–0,22	–0,08	–0,19	–0,11	0,08	0,61*	0,61*	0,48
Distance front block-starting line (a)	m	–0,27	0,36	0,63*	0,53*	0,72*	0,63*	0,31	0,05
Distance rear block-starting line (b)	m	–0,08	0,44	0,44	0,26	0,76*	0,63*	0,30	0,09
STARTING ACTION									
Start push-off angle (Kod)	°	0,31	0,33	0,30	0,20	–0,07	0,49	0,56	0,71*
Vertical start velocity (Vty)	ms^{-1}	0,40	0,47	0,40	0,57*	–0,36	0,22	0,41	0,42
Horizontal start velocity (Vtx)	ms^{-1}	–0,39	–0,57*	–0,60*	–0,66*	–0,03	–0,43	–0,49	–0,58
Angle between thighs (Kms)	°	–0,22	–0,17	0,02	–0,60	–0,12	–0,78*	–0,79*	–0,75*
Relative push-off force impulse of front foot (SIRSO)	Ns/kg	0,51	–0,22	–0,63*	–0,76*	–0,09	–0,07	–0,02	–0,34
Push-off force impulse of front foot (SISO)	Ns	0,49	–0,17	–0,57*	–0,71*	0,07	–0,01	0,00	–0,35
Maximal force gradient of front foot (SMGS)	N/s	0,38	–0,30	–0,71*	–0,78*	–0,06	–0,16	–0,06	–0,13
Start reaction time of front foot (SST)	s	0,43	–0,34	–0,74*	–0,66*	0,49	–0,58*	0,55	0,30
Maximal relative force gradient of front foot (SMGRS)	N/kg/s	0,37	–0,31	–0,71*	–0,76*	–0,23	–0,24	–0,10	–0,07
Maximal relative force of pressure of front foot (SMRSP)	N/kg	0,46	–0,35	–0,75*	–0,85*	–0,13	–0,03	0,10	–0,15
Maxsimal force of pressure of front foot (SMSP)	N	0,46	–0,32	–0,72*	–0,83*	0,05	0,03	0,11	–0,22
Average force gradient of front foot (SPGS)	N/s	0,50	–0,22	–0,64*	–0,81*	0,03	0,05	0,11	–0,12
Average relative force gradient of front foot (SPRGS)	N/kg/s	0,51	–0,25	–0,69*	–0,84*	–0,11	0,00	0,10	–0,08
Average relative force of pressure of front foot (SPRSP)	N/kg	0,45	–0,14	–0,49	–0,66*	–0,20	–0,20	–0,12	–0,43
Average force of pressure of front foot (SPSP)	N	0,42	–0,08	–0,41	–0,59*	0,16	–0,09	–0,07	–0,42
Latent reaction time of front foot (SRT)	s	0,77*	0,30	–0,28	–0,56*	0,20	0,21	0,46	0,48
Time to maximal force of front foot (STMAX)	s	0,69*	–0,55*	–0,55*	–0,66*	0,44	0,48	0,61*	0,56*
Latent reaction time of rear foot (ZRT)	s	0,86*	–0,19	–0,19	–0,47	0,16	0,17	0,46	0,33
Time to maximal force of rear foot (ZTMAX)	s	0,79*	–0,41	–0,41	–0,60*	0,58*	0,59*	0,69*	0,50*
START ACCELERATION – FIRST STRIDE									
Push-off angle (Kod 1)	°	0,37	0,45	0,41	0,22	–0,40	–0,24	–0,11	–0,10
Horizontal velocity (Vtx 1)	ms^{-1}	–0,10	–0,37	–0,44	–0,34	0,68*	–0,16	–0,47	–0,41
Vertical velocity (Vty 1)	ms^{-1}	–0,10	–0,36	–0,44	–0,33	0,66*	–0,19	–0,50	–0,43
Velocity resultant (Vtxy 1)	ms^{-1}	–0,10	–0,37	–0,44	–0,34	**0,66***	–0,19	–0,50	–0,43
START ACCELERATION – SECOND STRIDE									
Push-off angle (Kod 2)	°	0,47	0,42	0,34	0,22	–0,07	0,69*	0,74*	0,83*
Horizontal velocity (Vtx 2)	ms^{-1}	–0,24	–0,54	–0,58*	–0,39	0,27	–0,48	–0,69*	–0,74*
Velocity resultant (Vtxy 2)	ms^{-1}	–0,22	–0,52	–0,56*	–0,38	0,27	–0,49	–0,69*	–0,74*
Angle of velocity resultant (Ksv 2)	°	0,37	0,24	0,28	0,18	–0,43	–0,19	0,10	0,19

Legend:
U – measurement unit
C5 – time to 5 m
C10 – time to 10 m
C20 – time to 20 m
C30 – time to 30 m
*** – coefficient of correlation is statistically significant**

A generally low correlation can, however, be seen between the kinetic and kinematic variables with the initial phase of the start acceleration on 5 m, both for males as well as females. For males we find statistically significant correlation coefficients for the latent reaction time of the front and rear foot (r=.86, r=.69) and the time parameters of maximal force of the rear (r=.79) and front foot (r=.77). For the female sprinters the correlation coefficients show the starting action to be defined mostly by the position of the starting blocks and maximal force of the rear foot (r=.58). These variables also significantly correlate with the start acceleration on

10 m. For the males we find the start acceleration on 10 m to be correlated with just one parameter – horizontal start velocity (r=–.57).

In light of the level of correlation we can state that the start and the start acceleration on the distance of the first 10 m are two very different motor tasks, whose successfulness depends on very specific factors. The low correlation of the start with the start velocity in the first 10 m was also found by other authors (Sanderson, 1991; Delecluse, 1992). The starting action is an acyclic movement and obviously differs significantly from the viewpoint of biomechanical structure from the cyclic movement in the first phase of the start acceleration.

Specially noticeable is the positive correlation of the angle of the placement of the foot into the front block with the efficiency of the sprinter's acceleration at 20 m (r = .63). The set position, where the foot of the front leg is, in regard to the block, in dorsal flexion and ensures better conditions for the push-off from the front block. The effect is greater because of the excentric-concentric action of the extensors of the foot and the incorporation of elastic energy into the push-off impulse (Guissard, 1992). This phenomena was not noticed with female sprinters.

The kinetic parameters of the start are much more correlated with the start acceleration with male sprinters than female ones. The key role goes to the kinetic variables of the front foot, which are correlated especially with the times at 20 and 30 m. The biggest correlations are with: maximal and relative force of pressure (r = –.83, r = –.85), maximal force gradient (r = –.78), force impulse (r = –.71) and time to maximal force (r = .69).

CONCLUSION

The purpose of the study was to find, with proper measuring technology, those kinematic and kinetic parameters of the set position and starting action that generate the efficiency of the start acceleration, being one of the most important phases of top athletes in sprint. The following parameters best define an efficient start for both male and female sprinters: horizontal start velocity, start reaction time and impulse of push-off force from the front starting block. The postulated hypothesis H1 can therefore be partially confirmed. The results show that male sprinters differ from their female counterparts mostly in the kinetic and less in the kinematic variables. We were, however, unable to confirm hypothesis H2, where we supposed that the parameters of the set position and starting action decisively affect predominantly the initial phase of the start acceleration (0–5 m). On the basis of the results of this research we can state that the parameters of the start do effect the efficiency of the second phase of the start acceleration (20–30 m), both for male and female sprinters. This points to the fact that start acceleration immediately after the start has a very specific structure and is evidently independent of the kinematic and kinetic factors of the start.

REFERENCES

Bosco, C., Komi, P. V., & Ito, A. (1981). Prestretch potentiation of human skeletal muscle during ballistic movement. *Acta Physiologica Scandinavica, 111*, 135–140.

Bosco, C. (1985). Stretch-shortening cycle in skeletal muscle function and physiological considerations on explosive power in man. *Atleticastudi, 1*, 7–113.

Brüggemann, G. P. (1990). *Scientific Research Project at the Games of the XXXIV Olympiad Seoul 1988.* IAAF, Supplement.

Bührle, M., Schmidtbleicher, D., & Ressel, H. (1983). Die spezielle Diagnose der einzelnen Kraftkomponenten in Hochleistungssport. *Leistungssport, 3,* 11–16.

Coppenolle, H., & Delecluse, C. (1989). Technology and development of Speed. *Athletics Coach, 23 (1)*, 82–90.

Coppenolle, H., Delecluse, C., Goris, M., Diels. R., & Kraayenhof H. (1990). An evaluation of the starting action of world class female sprinters. *Track Technique, 90*, 3581–3582.

Delecluse, C., Coppenolle, H., & Goris, M. (1992). A model for the scientific preparation of high level sprinters. *New Studies in Athletics, 7 (4),* 57–64.

Delecluse, C., Coppenolle, H., Diels, R., & Goris, M. (1992). The F.A.S.T. project-A scientific follow-up of sprinting abilities. *New Studies in Athletics, 11 (2–3)*, 141–143.

Guissard, N., & Hainaut, K. (1992). EMG and mechanical changes during sprint start at different front block obliquities. *Medicine and Science in Sport and Exercise, 24 (11),* 1257–1263.

Hoster, M., (1981). Weg-, Zeit-, und Kraft-Parameter als Einflussgrossen beim Sprintstart in der Leichtathletik. *Leistungssport, 2*, 110–117.

Korchemny, R. (1992). A new concept for sprint start and acceleration training. *New Studies in Athletics, 7(4)*, 65–72.

McClements, J., Sanders, L., & Gander B. (1996). Kinetic and kinematic factors related to sprint starting as measured by Saskatchewan Sprint Start Team. *New Studies in Athletics, 11 (2–3)*, 133–135.

Mero, A. (1988). Force-Time Characteristics and Running Velocity of Male Sprinters During the Acceleration Phase of Sprinting. *Research Quarterly, 59, (2),* 94–98.

Mero, A., Luhtanen, P., & Komi, V. (1989). Segmentalle Krafterzeugung und Geschwindigkeit des Koerperschwerpunkts in den Kontaktphasen beim sprint. *Leistungssport, 4*, 35–39.

Moravec, P., Ruzicka, J., & Susanka, P., Dostal, E., Nosek, M. (1988). Time analysis of the 100 meters events at the II. World Championship in Athletics. *New Studies in Athletics, 3 (3),* 61–96.

Qing, L., & Kruger, A. (1995). Die Beschleunigungphase im Sprint ist nicht ein, sondern zweiteilig. *Leistungssport, 3,* 39–45.

Sanderson, L., McClements, J., & Gander, R. (1991). Development of apparatus to provide immediate accurate feedback to sprinters in the normal training environment. *New Studies in Athletics, 6 (2)*, 33–41.

Schmalz, T., & Turk-Noack, U. (1978). Untersuchung des Kraftverlaufs beim Tiefstart. *Leistungssport, 4*, 433–438.

Schot, P., & Knutzen, K. M. (1992). A Biomechanical Analysis of Four Sprint Start Positions. *Research Quarterly for Exercise and Sport, 63 (2),* 137–147.

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KINEMATISCHE UND KINETISCHE PARAMETER DES STARTS BEIM SPRINT UND DAS MODELL DER STARTBESCHLEUNIGUNG BEI SPITZENSPRINTERN

(Zusammenfassung des englischen Textes)

Die Forschung hatte zum Ziel, die wichtigsten kinematischen und kinetischen Parameter der Vorbereitungsstellung und des Starts und ihre Korrelation mit der Startbeschleunigung zu finden. Das Muster bestand aus dreizehn Männern und elf Frauen. Zur Aufzeichnung der kinematischen Startparameter und der Parameter der Startbeschleunigung wurde ein analytisches 2-Dimensionen-Videosystem (APAS) angewendet. Die kinematischen Parameter des Starts wurden mit Hilfe modifizierter Startblocks (MMIP) mit eingebauten Meßsensoren gemessen. Die Zeitparameter der Startbeschleunigung wurden mit Hilfe von vier 5–10–20–30 m vor der Startlinie angeordneten Paaren von Fotozellen gemessen. Die statistische Analyse wurde mit Hilfe des statistischen SPSS Päckchens durchgeführt.

Die Effektivität des Starts beim Sprint bei beiden Geschlechtern wurde durch folgende Größen definiert: horizontale Startgeschwindigkeit, Schwerpunkt, Zeitreaktion beim Start, Kraftimpuls und maximaler Kraftanstieg am vorderen Startblock. Wesentliche Unterschiede (p 0,05) zwischen den Geschlechtern treten vor allem im Bereich der kinetischen Parameter des Starts auf. Die höchste Korrelation hat der Block der kinetischen Parameter des Starts – maximale und relative Druckkraft, maximaler Kraftanstieg, Kraftimpuls und zur Erreichung der Maximalkraft notwendige Zeit; in den kinematischen Parametern ist es die horizontale Startgeschwindigkeit, der Schwerpunkt, und der Knöchelwinkel am vorderen Startblock. Frauen haben weitaus niedrigere Korrelationen, nur zwei bedeutende Koeffizienten wurden erworben – zur Erreichung der Maximalkraft notwendige Zeit am vorderen und hinteren Startblock. Die im Allgemeinen niedrigen Korrelationen zwischen dem Start und der Startbeschleunigung, besonders die ersten fünf Meter nach dem Start, sind Folge der Differenzen in der biomechanischen motorischen Struktur.

Schlüsselwörter: Sprinterstart, Startbeschleunigung, Kinematik, Kinetik, Spitzensprinter, Frauen, Männer.

KINEMATICKÉ A KINETICKÉ PARAMETRY STARTU PŘI SPRINTU A MODEL STARTOVNÍ AKCELERACE VRCHOLOVÝCH SPRINTERŮ

(Souhrn anglického textu)

Cílem výzkumu bylo nalézt nejdůležitější kinematické a kinetické parametry přípravné polohy a startu a jejich korelaci se startovní akcelerací. Vzorek sestával ze třinácti mužů a jedenácti žen. Pro záznam kinematických startovních parametrů a parametrů startovní akcelerace byl použit 2D analytický video systém (APAS). Kinetické parametry startu byly měřeny modifikovanými startovními bloky (MMIP) s vestavěnými měřicími senzory. Časové parametry startovní akcelerace byly měřeny čtyřmi páry fotobuněk (AMES) umístěnými 5–10–20–30 m od startovní čáry. Statistická analýza byla provedena pomocí statistického balíčku SPSS.

Efektivnost sprintérského startu u obou pohlaví byla vytvářena: horizontální startovní rychlostí těžiště, startovní časovou reakcí, silovým impulsem a maximálním silovým gradientem na předním startovním bloku. Významné rozdíly ($p < 0{,}05$) mezi pohlavími jsou především v oblasti kinetických parametrů startovní akce. Nejvyšší korelace startovní akcelerace u mužů má blok kinetických parametrů startovní akce – maximální a relativní síla tlaku, maximální silový gradient, silový impuls a doba potřebná k dosažení nejvyšší síly; z kinematických parametrů horizontální startovní rychlost těžiště a úhel kotníku na předním startovním bloku. Ženy mají mnohem nižší korelace, byly získány pouze dva významné koeficienty – doba potřebná k dosažení maximální síly na předním a zadním startovním bloku. Všeobecně nízké korelace mezi startem a startovní akcelerací, zvláště v prvních pěti metrech po startu jsou důsledkem rozdílů v biomechanické motorické struktuře.

Klíčová slova: sprinterský start, startovní akcelerace, kinematika, kinetika, vrcholoví sprinteři, muži, ženy.